



Unit 3 · Outcome 1 · Maintaining Biodiversity

Simpson's Index of Diversity

KK07

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 07 Simpson's Index of Diversity Counting species tells you how rich a place is

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



A COUNT: E.G. "12 SPECIES"

01 Three ways to measure species diversity

The study design names three tools. They answer different questions, and the exam expects you to know what each one can and can't see.

The total number of different species in a defined area. Count the species, ignore how many of each. Quick and intuitive.

A species is endemic if it's found naturally nowhere else on Earth. High endemism means a place holds irreplaceable diversity — losing it is a global, not local, loss.

A single number from 0 to 1 that combines richness and evenness — how the individuals are spread across the species.

◆ NOTE

► The distinction examiners reward Richness = how many species. Evenness = how balanced their abundances are. Endemism = how unique they are to this place. A site can be species-rich but very uneven (one species dominates), or species-poor but with high endemism. SID is the only one of the three that rolls richness and evenness into a comparable figure.

◆ ANALOGY

✧ Analogy — the music festival Two festivals each book the same 5 bands (equal richness). At Festival A the crowd is split evenly — 400 fans at each stage. At Festival B, 1,800 fans are jammed at one main stage and the other four play to 50 people each. Same line-up, completely different experience. Richness says they're identical; evenness — and therefore SID — knows they're not.

INTERACTIVE · EVENNESS RAISING THE CANOPY... FIVE SPECIES · 100 INDIVIDUALS · RICHNESS FIXED AT 5

02 Why a species count isn't enough

⚠ WATCH OUT

The takeaway Slide all the way to one dominant species and SID collapses toward 0 — even though richness is still 5. This is the heart of the dotpoint: richness is blind to evenness; SID is not. A monoculture (a wheat paddock, a lantana-choked gully) can have several species present yet near-zero diversity because one form runs the show.

03 The formula, decoded

◆ COMMAND WORDS

VCAA gives you the formula in the exam — your job is using it without slipping. Hover or tap each part to see what it means.

◆ NOTE

The fraction alone ($\sum(n-1)/N(N-1)$) This is D — the probability two random individuals are the same species (dominance). Big D = low diversity. Easy to stop here and report the wrong direction.

◆ NOTE

✓ Subtract from 1 → $SID = 1 - D$ = probability they're different species. Now big = diverse. The "1 -" is the single most-forgotten step on the exam.

◆ ANALOGY

✧ Why $n(n-1)$ and not just n^2 ? When you pick one individual, then pick a second without replacing the first, there are only $n-1$ of that species left out of $N-1$ total. So the chance of drawing the same species twice is $n(n-1)/N(N-1)$. It's sampling without replacement — exactly like drawing two cards from a deck without putting the first back.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

SITE A · INTACT GULLY LAYING OUT THE QUADRAT... EACH MARKER = ONE PLANT, COLOURED BY SPECIES. CLICK A SPECIES IN THE TABLE TO HIGHLIGHT IT.

04 Worked example — two plots at Kestrel Ridge

A Year 12 class surveys ground-layer plants in two 4 m^2 quadrats at the fictional Kestrel Ridge Reserve (box-ironbark woodland, central Victoria). Both quadrats hold the same 5 species and the same 56 individuals. Use the toggle to switch sites and watch the calculation rebuild.

Species	n	n - 1	n(n-1)
Totals	56	—	598

◆ NOTE

► Result Site A: $SID \approx 0.81$. The five species are fairly even, so two random plants are very likely to be different species — high diversity.

◆ NOTE

► The punchline Site A ≈ 0.81 , Site B ≈ 0.26 . Identical richness (5) and identical N (56) — yet SID separates them cleanly, because at the grazed edge a single grass dominates (48 of 56 individuals). This same-richness/different-SID setup is a classic VCAA question.

05 Reading the number

A bare " $SID = 0.81$ " earns little. Marks come from interpreting it in context and comparing.

SID near 0	Low diversity — one or few species dominate (monoculture, degraded site)
SID mid (~0.5)	Moderate — some spread but uneven
SID near 1	High diversity — many species, evenly abundant

A higher SID generally signals an ecosystem that is more resilient and more stable: more species and more even abundances mean redundancy in roles, so the system can absorb disturbance (disease, drought, fire) without collapse. Low SID communities are fragile — knock out the dominant species and the whole system wobbles.

◆ NOTE

Comparing sites — the sentence that scores "Site A (SID = 0.81) has a higher Simpson's Index of Diversity than Site B (SID = 0.26), indicating Site A is more diverse — both species are evenly represented rather than dominated by one — and therefore likely more resilient to disturbance." Note the structure: value → comparison → meaning → consequence.

⚠ WATCH OUT

Limits of SID — worth a mark SID is a number, not the whole story. It treats all species as equal — a weed and an endemic, threatened species count the same. A site overrun by invasive species can post a healthy SID. Pair SID with endemism and conservation status before judging a site's worth.

07 Command-word decoder

◆ COMMAND WORDS

The verb in the question tells you exactly what to deliver. Same topic, very different answers.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

08 Exam traps

⚠ EXAM TRAPS

The marks lost on this dotpoint are predictable. Open each to defuse it.

09 P.E.L writing trainer



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

P-E-L WRITING

For "explain" and "compare" answers, build each point as Point → Explain → Link. Step through a model response below.

10 Good vs weak — the same question, two answers

Q: "Sites A and B contain the same five species and the same number of individuals, yet Site A has a higher Simpson's Index of Diversity. Explain why." (3 marks)

• NOTE

Why it loses marks: the sites have equal richness, so "more species" is wrong and self-contradicting. "Better/worse/wins" isn't scientific language. No mention of evenness or what SID measures.

• NOTE

Why it scores: identifies equal richness, names evenness as the cause, explains the dominance in B, and links it back to what SID actually measures.

11 Practice & self-mark

★ PRACTICE & SELF-MARK

Five graded questions, 20 marks total. Attempt each on paper first, reveal the model answer, then award yourself marks — the dock tracks your total.

◆ NOTE

- How to self-mark honestly Award a mark only if your wording carries the same idea as a bolded mark-point — not just a near-miss. For calculations, full marks need the working and the final SID with the "1 -" applied.

12 Key takeaways

◆ KEY TAKEAWAY

Three measures: species richness (how many species), endemism (found only here), and SID (richness + evenness in one 0–1 number). ◆ $SID = 1 - \sum n(n-1)/N(N-1)$, where n = individuals of each species and N = total individuals. The "1 –" makes higher = more diverse. ◆ It's a probability: the chance two random individuals are different species. ◆ Evenness is the point: same richness can give very different SID. Dominance by one species drives SID toward 0. ◆ Interpret, don't just state: value → compare → meaning (diversity) → consequence (resilience/stability). ◆ Know its blind spot: SID can't tell a weed from an endemic; pair it with endemism and conservation status.

UNIT 3 · OUTCOME 1 · KEY KNOWLEDGE 07

Simpson's Index of Diversity

Counting species tells you how rich a place is. But two forests with the same species list can be wildly different. Simpson's Index of Diversity captures what a headcount misses — evenness — by asking one elegant question: pick two individuals at random; how likely is it they're different species?

◆ NOTE

Where this sits in the chapter You've sampled the woodland with quadrats and transects (KK06) and you've got raw counts. This dotpoint is what you do with those numbers: turn a table of tallies into a single, comparable measure of diversity — and defend what it means.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Site A has more species so it is more diverse. It has a higher number which means it is better. Site B is worse because it has less. So A wins. Why it loses marks: the sites have equal richness, so "more species" is wrong and self-contradicting. "Better/worse/wins" isn't scientific language. No mention of evenness or what SID measures.

✓ STRONG / MODEL ANSWER

Both sites have the same species richness (5) and the same total individuals, so the difference must come from evenness. In Site A the individuals are spread evenly across the five species, whereas in Site B one species dominates the count. Because Simpson's Index of Diversity rewards an even spread, two randomly chosen individuals in Site A are more likely to be different species, giving Site A the higher SID and indicating greater diversity. Why it scores: identifies equal richness, names evenness as the cause, explains the dominance in B, and links it back to what SID actually measures.

